Activity 4- Data bias and exclusion

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Question 1: Thinking about potential bias and exclusion is the data collected in such a way that certain groups could be underrepresented or completely absent from the dataset?

Answer:

- **Online Survey**:
  - We are told the "… agency collected online survey responses from users…"
  - Conducting the survey online by its very nature is going to exclude sections of the population who will not be able to access it for a variety of reasons.
- **Age and technical ability breakdown:**
  - Age per Digital Self-Efficacy (DSE) scale (Ulfert-Blank and Schmidt, 2022).

=== AGE per Digital Self-Efficacy (DSE) value ===

Tech Conf	min	max	mean	median	count
DSE1	47	62	55.2	55.0	5
DSE2	33	60	48.2	50.0	4
DSE3	25	36	31.7	34.0	3
DSE4	29	44	38.2	40.0	4
DSE5	23	31	27.2	27.5	4

Ulfert-Blank, A.-S., & Schmidt, I. (2022). *Digital Self-Efficacy Scale*. APA PsycTests. https://doi.org/10.1037/t87118-000

  - People with lower DSE score have lower confidence in their abilities i.e. DSE 1 and 2, while people of younger ages have higher scores DSE 4 and 5.
  - The age range is >= 23 and <=62
    - There are no pensioners or young adults/teenagers present in the dataset.
- **Geographical spread – rural/urban divide:**

=== Locations ===

City	
London	6
Manchester	4
Leeds	3
Bristol	2
Birmingham	2
Sheffield	2
Liverpool	1

  - There is no representation for rural areas.
- **Other classifications groupings - i.e. Gender / social status / spoken languages**
  - We have no indication who took the survey from these groupings, their usage behaviour and requirements may differ.

- **Representative population sample:**
    - Overall, the dataset does not look to be a representative population sample

Question 2: Look at the "HasDisability" column. What do you notice about how disability is represented?

Answer:

- **Disclosure**
    - Nobody has explicitly said they have a disability. This could be for a variety of reasons:
        - "Prefer not to say" answers could suggest they do not trust the data collector or how their data may be used. Some respondents may not feel comfortable answering this question honestly or at all.
        - If it were a representative population sample how many would we expect to find with disabilities? We would expect to see some respond "yes".
- **Definition of Disability**
    - There is no definition of disability or means to grade or assess oneself.
    - There can be subjectivity in terms of how someone views themselves compared to others in relation to disability.
    - How do people classify themselves? A "**yes**" or "**no**" answer may not be the most appropriate answer options to present to users for this question.

Question 3: Why might people with disabilities not appear at all in this dataset?

Answer:

- **Did not hear about it**
    - Disabled people may not have been aware of it
- **Unable to access survey**
    - Disabled people may not be able to access or complete the survey due to their disability, i.e. manual dexterity limitations for example. By its nature anyone who cannot access and complete an online survey will not be in the dataset.

- **Legal obligation**
    - Data Agency may have decided consciously to not collect "special category" data in any meaningful way to avoid any legal obligations around the collection, storage, usage and maintenance that would come with that data.
    - https://www.citizensinformation.ie/en/government_in_ireland/data_protection/overview_of_general_data_protection_regulation.html

- **Trust**
    - Lack of trust in how the data will be used may be preventing people from answering accurately, there may be privacy concerns.

Question 4: Describe one reason why a public agency might intentionally choose not to collect disability data.

Answer:

- **Difficulty gaining approval and legal protections**
    - There can be additional work and difficulties gaining approval from a Research Ethics Committee to include users with disabilities in research due to issues concerning informed consent and protection from adverse practices. This coupled with the additional legal responsibilities that come with such special category data under GDPR legislation may mean some research bodies may intentionally exclude them.

Question 5: What recommendation would you give for a more inclusive approach to data collection?

Answer:

- **Co-Design** – people with "**lived experience**" - allow participants or representative bodies design the survey and how the survey is conducted, i.e. the where, when, how
- **Active Inclusion** – pro-actively identify minority groups and try to engage with them. i.e. people who likely have low DSE scores, pensioners, young adults, teens etc. (Source: WEF, 2018)
- Encourage **Reflexivity** within the research group to think abuot how the information in the dataset represents the respondents, not to just blindly eliminate outliers.
- Better definition of disability types, the current term is such a broad definition its actual usage is very limited in terms of what you can infer from it i.e. mobility, independence, level and types of assistance that may be required, etc
- Clearly define and communicate assurances around the data usage, Anonymisation vs Pseudonymisation, GDPR adherence etc
- Improve the definition of disability perhaps including details of types with a Lickert type scale, or based around data from Irish Human Rights and Equality Commissin (IHREC)
- Make it easier or have alternative completion methods for DSE 1 and DSE2 users, carry out accessibility testing prior to rollout.

References

- (Ulfert-Blank and Schmidt, 2022).
- Ulfert-Blank, A.-S., & Schmidt, I. (2022). *Digital Self-Efficacy Scale*. APA PsycTests.
    - https://doi.org/10.1037/t87118-000
- https://www.citizensinformation.ie/en/government_in_ireland/data_protection/overview_of_general_data_protection_regulation.html
- Irish Human Rights and Equality Commissin (IHREC)
- (Source: WEF, 2018)